

Investment Memorandum

Quantitative Deep-Tech Investment Screening Framework

Recommendation: Allocate \$1,000,000 to SpaceX

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Executive Summary

This memorandum presents a quantitative investment screening framework developed to identify the strongest candidate for a hypothetical \$1 million investment across 49 leading deep-tech companies spanning Artificial Intelligence, Robotics, Defence Technology, Space Technology, and Quantum Computing. Because these companies differ significantly in business maturity, financial transparency, and commercialization stage, a single financial metric is insufficient for fair comparison. To address this challenge, I developed an evaluation methodology based on 31 financial, technological, and strategic criteria.

The framework integrates Weighted Scoring, AHP, and TOPSIS, followed by Monte Carlo simulation, sensitivity analysis, and qualitative investment committee review to ensure the recommendation remains robust under different assumptions.

Across all evaluation methods, SpaceX consistently ranked first, achieving the highest weighted score (85.85), the highest TOPSIS score (0.703), and retaining Rank 1 across all weighting strategies. Monte Carlo analysis further showed a 64.9% probability of maintaining the top position and a 91.5% probability of remaining within the Top 3. Based on both quantitative evidence and qualitative due diligence, SpaceX represents the strongest long-term investment opportunity within the evaluated universe.

1 Investment Objective

The objective was to identify the strongest long-term investment opportunity rather than the most recognizable company. To achieve this, 49 deep-tech companies were evaluated using a common framework combining financial metrics with technology leadership, commercialization, execution capability, scalability, competitive positioning, and investment risk. The recommendation is therefore based on balanced performance across multiple independent criteria instead of exceptional performance in any single metric.

Investment Snapshot

Investment Universe	49 Deep-Tech Companies
Strategic Sectors	Artificial Intelligence, Robotics, Defence, Space Technology, Quantum Computing
Evaluation Criteria	31 Quantitative and Qualitative Features
Decision Methods	Weighted Scoring, AHP, TOPSIS
Validation Methods	Monte Carlo Simulation + Sensitivity Analysis
Final Recommendation	SpaceX
Investment Allocation	\$1,000,000 (Hypothetical)
Recommendation Strength	High Conviction Buy
Final Rank	1 / 49

2 Investment Decision Framework

The investment recommendation was generated through a structured screening process designed to evaluate both public and private deep-tech companies using a common decision framework. Since these companies differ substantially in financial transparency, business maturity, and commercialization stage,

relying on a single evaluation metric would produce biased comparisons. The proposed framework integrates feature engineering, Weighted Scoring, AHP, TOPSIS, Monte Carlo robustness analysis, sensitivity testing, and qualitative investment committee review to produce a transparent and evidence-based investment recommendation.

3 Quantitative Results

The framework consistently identified SpaceX as the strongest investment opportunity across all quantitative evaluation methods. Rather than excelling in a single metric, SpaceX achieved consistently high performance across technology leadership, commercialization, execution capability, competitive positioning, and long-term scalability.

Table 1: Top Five Investment Opportunities

Rank	Company	Weighted Score	TOPSIS
1	SpaceX	85.85	0.703
2	Anduril Industries	84.91	0.691
3	OpenAI	84.36	0.684
4	Anthropic	83.92	0.678
5	Figure AI	83.51	0.671

The top five companies achieved relatively similar composite scores, indicating a highly competitive investment universe. SpaceX secured the highest overall ranking by maintaining consistently strong performance across technology leadership, commercialization, execution capability, competitive positioning, and scalability rather than relying on exceptional performance in a single category.

4 Investment Validation

To verify that the recommendation was not driven by a particular weighting strategy, Monte Carlo simulation and sensitivity analysis were performed.

Table 2: Robustness Summary

Metric	Result
Probability of Rank 1	64.9%
Probability of Top 3	91.5%
Probability of Top 5	96.5%
Median Rank	1

Table 3: Sensitivity Analysis

Weighting Method	Final Rank
Investor	1
AHP	1
Entropy	1
Equal Weight	1

Across all weighting strategies, SpaceX retained Rank 1, indicating that the recommendation remains stable under reasonable variations in evaluation assumptions.

5 Investment Committee Assessment

Dimension	Assessment
Technology	Reusable launch systems, Starlink ecosystem and vertical integration provide a durable competitive advantage.
Commercialization	Strong launch business, recurring satellite revenue and long-term government partnerships.
Growth Drivers	Starship development, Starlink expansion and increasing commercial space demand.
Risks	Private market liquidity, execution risk and regulatory uncertainty.
Verdict	Quantitative and qualitative analyses independently support SpaceX as the highest-conviction investment.

The qualitative assessment was consistent with the quantitative ranking, reinforcing confidence that the recommendation remained robust when strategic business factors were considered alongside numerical evaluation.

6 Final Recommendation

The investment recommendation is supported by four independent lines of evidence: quantitative ranking, ranking stability, sensitivity analysis, and qualitative due diligence. Together, these analyses consistently identify SpaceX as the strongest investment opportunity within the evaluated universe, providing a transparent and evidence-based justification for the proposed capital allocation.

Investment Decision

Allocate \$1,000,000 to SpaceX
High Conviction Recommendation

References

1. SpaceX. Official Company Website. <https://www.spacex.com>
2. Yahoo Finance. Historical financial market data.
3. Official company websites, annual reports, investor presentations, and publicly available corporate disclosures.
4. GitHub Repository (Project Source Code):
<https://github.com/DebadattaLiku/DeepTech-Investment-Framework>